Response to Reviewers' Comments

Journal: [Journal name]
Manuscript ID: [Manuscript ID]
Manuscript title: [Manuscript title]

Dear Editor and Reviewers,

Thank you for your time and constructive comments on our manuscript. We provide a point-by-point response below, explaining the changes made and identifying their locations in the revised manuscript.

[Describe how revisions are marked in the manuscript, if applicable. State whether page and line numbers refer to the clean or marked version.]

The main revisions are summarized as follows:

e first major revision.

econd major revision.

nove items as needed.

Thank you for considering our revised manuscript.

Yours sincerely,

[Corresponding author's name]
On behalf of all authors

Comments by the Editor

Comment 1:

[Insert the editor's comment verbatim.]

Response:

[Provide a direct response and describe the action taken. If no change was made, explain the reason with supporting evidence.]

Location in the revised manuscript: [Section, page, and line numbers, or not applicable.]

Revised text:

[Insert the relevant revised passage, if applicable.]

Comments by Reviewer 1

Comment 1:

The adversary is assumed to reconstruct the transmit signals of individual APs from its received observations. Since the received signal contains a superposition of multiple AP transmissions, it is unclear how these components can be reliably separated, especially in the presence of AN and imperfect CSI.

Response:

We thank the reviewer for this important comment and agree that the separation step requires an explicit treatment. Since all APs transmit common symbols, the aggregate observation $\mathbf{y}_a[n] = \sum_k \mathbf{H}_{a,k}\mathbf{x}_k[n] + \mathbf{n}_a[n]$ provides only M_{UE} observation dimensions per snapshot for $N_{\mathrm{tx}}M_{\mathrm{tx}}$ per-AP transmit dimensions. Blind separation of the per-AP components from $\mathbf{y}_a[n]$ alone is therefore not possible in general, and we do not claim otherwise. The original manuscript did not state clearly that this step is an assumption, and we have revised the threat model accordingly.

Following the threat model of [12], the revised manuscript explicitly adopts a *genie-aided strong-adversary model*, in which the per-AP components of the received samples are made available to the adversary. We adopt this model deliberately, for the following reasons.

1. *Conservative privacy assessment.* The genie removes the inter-AP superposition, which is the main obstacle for a practical adversary seeking AP-dependent directions. The privacy of the proposed design is therefore assessed against an adversary that is granted a capability that a practical adversary does not have in general.
2. *Plausibility.* Partial per-AP separation may become available in practice, e.g., when AP-specific transmissions are resolvable in time or frequency, or when the propagation delays of the distributed APs are resolvable over a wide bandwidth.
3. *Design principle.* The network cannot verify the receiver capability of a served user. A privacy mechanism should therefore not rely on the adversary being unable to separate the per-AP signals.

Regarding the AN, it is intentionally contained in the per-AP components available to the adversary. Keeping the AN visible in the raw observations, while nulling it at the legitimate combiner outputs, is the receive-processing asymmetry on which the proposed design relies. Regarding imperfect CSI, the adversary's channel-estimation error enters the residual term $\widetilde{\mathbf{n}}_{a,k}^{\mathrm{adv}}[n]$ of the signal model. Consistent with the strong-adversary model, the simulated adversary is further given accurate knowledge of its per-AP channels $\mathbf{H}_{a,k}$, which is now stated in Section IV. The identifiability of the per-AP reconstruction step is discussed in our response to Reviewer 2, Comment 2.

Location in the revised manuscript: Section II, threat-model paragraph; Section IV-A, simulation setup; Section IV-B, first paragraph. [Add page and line numbers.]

Revised text:

Section II: "Since all APs transmit common symbols, $\mathbf{y}_a[n]$ offers only M_{UE} dimensions for $N_{\mathrm{tx}}M_{\mathrm{tx}}$ per-AP transmit dimensions, so its per-AP decomposition is not identifiable from $\mathbf{y}_a[n]$ alone in general. Following [12], we therefore adopt a genie-aided strong-adversary model in which the per-AP components of the received samples are available to the adversary, as may occur when AP-specific transmissions are resolvable in time, frequency, or delay. Since the network cannot verify the receiver capability of a served user, this conservative model avoids relying on the adversary being unable to separate them."

Section IV-A: "Throughout, the beams are designed from LMMSE channel estimates, whereas the realized SINRs are evaluated on the actual channels, and the genie-aided adversary of Section II is further given accurate knowledge of $\mathbf{H}_{a,k}$."

Section IV-B: "Since the adversary is genie-aided, these success rates represent the leakage under a conservative attack model."

Comment 2:

Constraint (9g) does not necessarily guarantee that the fake direction becomes the dominant peak of the reconstructed beampattern. Moreover, the condition used to derive (10) is not explicitly enforced in the optimization.

Response:

We thank the reviewer for this careful observation and agree with both points. Under the ideal covariance decomposition, the reconstructed true-direction response is $\widehat{B}_k(\phi_k) = S_k + T_k$, where $S_k = \|\mathbf{a}^H(\phi_k)\mathbf{W}_{-a,k}\|_F^2$ and $T_k = |\mathbf{a}^H(\phi_k)\mathbf{z}_k|^2$. The original constraint (9g) bounds S_k through the fake-direction response $F_k = |\mathbf{a}^H(\phi_{\mathrm{fake},k})\mathbf{z}_k|^2$, but places no restriction on T_k, i.e., on the AN response in the true direction. The fake-direction response could therefore fall below the true-direction response even when (9g) holds, and the condition $T_k/F_k \le 1/M_{\mathrm{tx}}$ used to derive (10) was indeed not enforced. We have revised the design as follows.

(i) Revised fake-peak constraint. We now include the true-direction AN response in the constraint itself:

$$\left|\mathbf{a}^H(\phi_{\mathrm{fake},k})\mathbf{z}_k\right|^2 \ge \rho \left\|\mathbf{a}^H(\phi_k)\mathbf{W}_k\right\|_F^2 + \left|\mathbf{a}^H(\phi_k)\mathbf{z}_k\right|^2, \quad k \in \mathcal{K}_{\mathrm{res}}. \tag{9g$'$}$$

Since the remaining terms of $\widehat{B}_k(\phi_{\mathrm{fake},k})$ are nonnegative, $\widehat{B}_k(\phi_{\mathrm{fake},k}) \ge F_k$, and since $S_k \le \|\mathbf{a}^H(\phi_k)\mathbf{W}_k\|_F^2$, (9g$'$) directly gives

$$\widehat{B}_k(\phi_{\mathrm{fake},k}) - \widehat{B}_k(\phi_k) \ge F_k - S_k - T_k \ge (\rho - 1)\left\|\mathbf{a}^H(\phi_k)\mathbf{W}_k\right\|_F^2. \tag{10$'$}$$

Hence, $\rho > 1$ is sufficient for the fake-direction response to exceed the true-direction response, and this guarantee now follows from a constraint that the optimization enforces, without any auxiliary condition. The previous sufficient condition $\rho > M_{\mathrm{tx}}/(M_{\mathrm{tx}} - 1)$ is replaced by $\rho > 1$.

(ii) Tractability. In the CCCP update, (9g$'$) becomes $\rho\,\mathrm{Tr}(\mathbf{W}_k^H\mathbf{A}_k\mathbf{W}_k) + \mathbf{z}_k^H\mathbf{A}_k\mathbf{z}_k - \widetilde{\zeta}_k \le 0$, where $\widetilde{\zeta}_k$ is the affine lower bound of F_k. This is a convex quadratic constraint. Since $\widetilde{\zeta}_k \le F_k$ globally, every point feasible for the surrogate is feasible for (9g$'$), so the feasibility, monotonicity, and convergence arguments of Section III-B carry over unchanged.

(iii) Resolvability. The comparison in (10$'$) is meaningful only when the M_{tx}-element array can resolve the true and fake directions. We now make this explicit by imposing (9g$'$) on

$\mathcal{K}_{\text{res}} = \{k : |\sin\phi_k - \sin\phi_{\text{fake},k}| \geq 2/M_{\text{tx}}\}$, i.e., on the APs whose two directions are at least one Rayleigh width apart. In the default geometry, the target and the virtual location are nearly collinear with three transmit APs, for which $|\sin\phi_k - \sin\phi_{\text{fake},k}| \approx 0.03$–$0.04 < 2/M_{\text{tx}} = 0.125$. At these APs no distinguishable fake peak can be formed, and imposing the constraint would only consume sensing power. The remaining five APs satisfy the condition.

(iv) From the true direction to the dominant peak. We agree that (10′) compares the fake direction with the true direction only, and that it holds for the ideal covariance. It does not guarantee that $\phi_{\text{fake},k}$ is the global maximum over all θ, since the beams of the other users produce responses in their own directions, and the finite number of snapshots and the EM reconstruction error add residual cross terms. We have therefore revised the manuscript to claim only what is guaranteed, namely that the fake-direction response exceeds the true-direction response, and we evaluate the location of the reconstructed dominant peak numerically. For localization deception, the relevant event is that the dominant peak is displaced from the true direction, since a peak elsewhere also removes that AP's direction line from the target cell. At $\eta \approx 0.50$, over the APs in $\mathcal{K}_{\text{res}}$, the reconstructed fake-direction response exceeds the true-direction response in $[X]\%$ of the cases, and the dominant peak lies at the fake direction, the true direction, and elsewhere in $[X]\%$, $[X]\%$, and $[X]\%$ of the cases, respectively.

(v) Updated results. All results have been recomputed with (9g′). At $\eta \approx [X]$, the proposed design now reduces the localization success rate to $[X]\%$, with a sensing SNR of $[X]$ dB.

Location in the revised manuscript: Section III-B, problem (9) and the paragraph containing (10); the CCCP update (12); Section IV-B, first paragraph. [Add page and line numbers.]

Revised text:

Section III-B, constraint (9g):

$$\left|\mathbf{a}^H(\phi_{\text{fake},k})\mathbf{z}_k\right|^2 \geq \rho \left\|\mathbf{a}^H(\phi_k)\mathbf{W}_k\right\|_F^2 + \left|\mathbf{a}^H(\phi_k)\mathbf{z}_k\right|^2, \ \forall k \in \mathcal{K}_{\text{res}}.$$

Section III-B, paragraph containing (10): "With $\mathbf{W}_{-a,k}$ denoting $\mathbf{W}_k$ without $\mathbf{w}_{a,k}$, define $S_k = \|\mathbf{a}^H(\phi_k)\mathbf{W}_{-a,k}\|_F^2$, $F_k = |\mathbf{a}^H(\phi_{\text{fake},k})\mathbf{z}_k|^2$, and $T_k = |\mathbf{a}^H(\phi_k)\mathbf{z}_k|^2$. The ideal covariance decomposition gives $\widehat{B}_k(\phi_{\text{fake},k}) \geq F_k$ and $\widehat{B}_k(\phi_k) = S_k + T_k$, so (9g) yields

$$\widehat{B}_k(\phi_{\text{fake},k}) - \widehat{B}_k(\phi_k) \geq F_k - S_k - T_k \geq (\rho - 1)\left\|\mathbf{a}^H(\phi_k)\mathbf{W}_k\right\|_F^2.$$

Hence, $\rho > 1$ is sufficient for the fake-direction response to exceed the true-direction response. Since this comparison requires resolvable directions, (9g) is imposed on $\mathcal{K}_{\text{res}} = \{k : |\sin\phi_k - \sin\phi_{\text{fake},k}| \geq 2/M_{\text{tx}}\}$. The guarantee holds for the ideal covariance and relative to the true direction; the location of the reconstructed dominant peak under finite snapshots and reconstruction errors is evaluated in Section IV."

Section III-B, CCCP update (12), last constraint: "$\rho\,\text{Tr}(\mathbf{W}_k^H\mathbf{A}_k\mathbf{W}_k) + \mathbf{z}_k^H\mathbf{A}_k\mathbf{z}_k - \widetilde{\zeta}_k \leq 0, \ \forall k \in \mathcal{K}_{\text{res}}$."

Section IV-B: "By (10), $\rho > 1$ is sufficient for the fake-direction response to exceed the true-direction response at the $|\mathcal{K}_{\text{res}}| = 5$ APs with resolvable directions."

Comment 3:

[Insert the reviewer's third comment verbatim.]

Response:

[Provide the response and describe any resulting changes.]

Location in the revised manuscript: [Section, page, and line numbers, or not applicable.]

Revised text:

[Insert the relevant revised passage, if applicable.]

Comment 4:

The sensing SNR expression in (2) needs further clarification. The contributions from different APs are added without considering possible coherent terms. In addition, the AN waveform is assumed to be known at the legitimate sensing receivers, but its contribution is excluded from the sensing SNR.

Response:

We thank the reviewer for this comment. We have revised the sensing model in Section II to state the receiver processing and the assumptions behind (2) explicitly, and we address the two points below.

Coherent cross terms. Since all transmit APs radiate common symbols, the echo received at receive AP r is the superposition $\sum_k \alpha_{r,k}\sqrt{g_{r,k}}\,\mathbf{a}^H(\phi_k)\mathbf{W}_k\mathbf{s}[n]$, where $\mathbf{s}[n]$ collects the symbols and $\alpha_{r,k}$ is the complex bistatic echo coefficient with $\mathbb{E}\{|\alpha_{r,k}|^2\} = \xi_{r,k}^2$. Its instantaneous power therefore contains cross terms between transmit APs, whose values depend on the relative phases of $\{\alpha_{r,k}\}_k$. These phases are not known to the network, because the target position is known only coarsely relative to the carrier wavelength ($\lambda_c \approx 15.8$ cm at 1.9 GHz) and the target response varies with the aspect angle. We therefore model them as independent and uniformly distributed across transmit–receive AP pairs. Under this model, the cross terms vanish in expectation, and (2) is the per-snapshot sensing SNR averaged over the echo phases, which is why it is referred to as the *average* sensing SNR. The revised manuscript now states this assumption explicitly. A design that exploits coherent superposition at the target would require phase-accurate knowledge of the target position and phase synchronization of the transmit APs at the target, which we do not assume for the sensing function.

AN contribution. The reviewer is correct that the AN echo was excluded from (2) without explanation. Since $s_z[n]$ is known at the receive APs and independent of the data and sensing symbols, its echo is separated by matched filtering and could in principle be exploited. We keep it out of the design objective deliberately. The AN echo from the true target is proportional to $T_k = |\mathbf{a}^H(\phi_k)\mathbf{z}_k|^2$, so including it in the objective would reward AN power in the true direction. This would raise the true-direction peak of the beampattern reconstructed by the adversary and work against the fake-peak constraint. Exploiting the AN echo at the receivers after the design would add $\sum_k c_k T_k \geq 0$ to γ_t. Hence, the reported γ_t is a lower bound on the achievable sensing SNR, and the reported 3.6 dB loss is an upper bound on the sensing cost of the proposed design. Both points are now stated in Sections II and III.

Location in the revised manuscript: Section II, sensing-model paragraph; Section III-B, paragraph following (10). [Add page and line numbers.]

Revised text:

> *Section II:* "Each receive AP employs M_{rx} antennas, steers its receive beam toward the target, and matched-filters its received samples with the known transmitted symbols, where the data symbols, being available at the central processor, act as additional probing signals. The bistatic echo phases are not known to the network to within the carrier wavelength and vary with the aspect angle, and are therefore modeled as independent and uniformly distributed across transmit–receive AP pairs. Averaging over these phases removes the cross terms between the echoes of different transmit APs. With $\mathbf{W}_k \triangleq [\mathbf{w}_{1,k}, \ldots, \mathbf{w}_{N_{\mathrm{UE}},k}, \mathbf{w}_{t,k}]$, the per-snapshot sensing SNR, averaged over the echo phases and the N_{rx} receive APs, is [(2), unchanged]."
>
> *Section III-B:* "Since $s_z[n]$ is known at the receive APs and independent of the other symbols, matched filtering separates its echo. This echo is excluded from (2) and from the design objective, so that the objective does not reward AN power in the true direction, which would raise T_k. Exploiting it at the receivers would add $\sum_k c_k T_k \geq 0$, so γ_t is a lower bound on the achievable sensing SNR."

Comments by Reviewer 2

Comment 1:

There is a potential inconsistency in the sensing-cost model. Equation (2) defines the sensing SNR using only the communication/sensing beam matrix $\mathbf{W_k}$ and excludes the AN beam $\mathbf{z_k}$, while the manuscript later states that $s_z[n]$ is known at the sensing receivers and that its echo can be separated from $s_t[n]$ by ideal matched filtering. If $s_z[n]$ is known, its target echo is itself a known probing signal and can, in principle, contribute additional sensing information; if it is deliberately discarded, the reported 3.6 dB sensing-SNR loss is specific to that receiver definition. Can the sensing signal model explicitly state whether the AN echo is exploited, discarded, or treated as interference?

Response:

We thank the reviewer for pointing out this inconsistency. The reviewer is correct that the known AN echo is itself a probing signal, and that the original manuscript did not state how it is handled. The revised sensing model now states explicitly that the AN echo is *separated and discarded*: since $s_z[n]$ is known at the receive APs and independent of the data and sensing symbols, matched filtering separates its echo, and it is neither exploited in (2) nor treated as interference. The residual cross-correlation between independent unit-power symbols after matched filtering decays as $1/N$ in power and is neglected for $N = 128$.

We discard the AN echo in the design deliberately, rather than for convenience. The AN echo from the true target is proportional to $T_k = |\mathbf{a}^H(\phi_k)\mathbf{z}_k|^2$. Including it in the objective would therefore reward AN power in the true direction, which raises the true-direction peak of the beampattern reconstructed by the adversary and works against the fake-peak constraint (9g).

We also agree that the 3.6 dB loss is specific to this receiver definition, and we have qualified it accordingly. If the receivers additionally exploit the AN echo, the sensing SNR becomes $\gamma_t + \sum_k c_k T_k \geq \gamma_t$. The reported γ_t is therefore a lower bound on the achievable sensing SNR, and the reported 3.6 dB loss is an upper bound on the sensing cost of the proposed design. In addition, our response to Reviewer 3, Comment 2, translates this loss into detection and false-alarm probabilities.

Location in the revised manuscript: Section II, sensing-model paragraph; Section III-B, paragraph following (10); Section IV-B, discussion of Fig. 3. [Add page and line numbers.]

Revised text:

> *Section III-B:* "Since $s_z[n]$ is known at the receive APs and independent of the other symbols, matched filtering separates its echo. This echo is excluded from (2) and from the design objective, so that the objective does not reward AN power in the true direction, which would raise T_k. Exploiting it at the receivers would add $\sum_k c_k T_k \geq 0$, so γ_t is a lower bound on the achievable sensing SNR."
>
> *Section IV-B:* "Since the AN echo is discarded in (2), this loss is an upper bound on the sensing cost of the proposed design."

Comment 2:

The entire threat model relies on the assumption that the adversarial user can separate its received superposition into per-AP raw components and then reconstruct $\hat{\mathbf{x}}_k[n]$ for each AP. However, the actual observation is $\mathbf{y}_a[n] = \sum_k \mathbf{H}a, k\mathbf{x}_k[n] + \mathbf{n}_a[n]$, while each $\mathbf{x}_k[n] \in \mathbb{C}^{M_{\text{tx}}}$ is unknown; with $N_{\text{tx}}M_{\text{tx}}$ transmit dimensions but only M_{UE} received dimensions per snapshot, this decomposition is generally underdetermined. What identifiability conditions make the assumed EM reconstruction of each AP waveform possible, particularly for the default setting $N_{\text{tx}} = 8$, $M_{\text{tx}} = 16$, and $M_{\text{UE}} = 16$?

Response:

We thank the reviewer for this insightful comment and fully agree with the dimension count. Since the APs transmit common symbols, $\mathbf{y}_a[n]$ offers $M_{\text{UE}} = 16$ observation dimensions per snapshot, against $N_{\text{tx}}M_{\text{tx}} = 128$ per-AP transmit dimensions in the default setting. The decomposition of $\mathbf{y}_a[n]$ into per-AP components is therefore underdetermined, and the original manuscript did not state clearly that this step is an assumption rather than a consequence of the signal model. We have revised the threat model to separate the attack into two stages and to state the identifiability status of each.

(i) Per-AP separation (assumed). The separation of $\mathbf{y}_a[n]$ into per-AP components is not identifiable from $\mathbf{y}_a[n]$ alone in general. Following [12], we now state explicitly that we adopt a genie-aided strong-adversary model, in which the per-AP observations $\widetilde{\mathbf{y}}_{a,k}[n]$ are available to the adversary. The genie removes the inter-AP superposition, which is the main obstacle for a practical adversary, so the privacy of the proposed design is assessed against an adversary that is granted a capability that a practical adversary does not have in general. Since the network cannot verify the separation capability of a served user, we consider it appropriate that the defense does not rely on this capability being limited. Partial separation may become available in practice, e.g., when AP-specific transmissions are resolvable, or under wideband operation with resolvable inter-AP delays.

(ii) Per-AP reconstruction (identifiable under a stated condition). Given $\widetilde{\mathbf{y}}_{a,k}[n] = \widehat{\mathbf{H}}^{\text{adv}}_{a,k}\widetilde{\mathbf{x}}_k[n] + \widetilde{\mathbf{n}}^{\text{adv}}_{a,k}[n]$, the recovery of $\widetilde{\mathbf{x}}_k[n]$ is a linear inverse problem with M_{UE} equations and M_{tx} unknowns. It is identifiable when $\widehat{\mathbf{H}}^{\text{adv}}_{a,k}$ has full column rank, which holds with probability one under Rayleigh fading whenever $M_{\text{UE}} \geq M_{\text{tx}}$. The default setting satisfies this condition with $M_{\text{UE}} = M_{\text{tx}} = 16$. Moreover, since the beampattern in (3) depends only on the covariance $\mathbf{R}_{\widetilde{\mathbf{x}}_k}$, whose rank is at most $N_{\text{UE}} + 1$ (the other users' streams, the sensing stream, and the AN stream), a weaker sufficient condition is that $\widehat{\mathbf{H}}^{\text{adv}}_{a,k}$ is injective on the column space of $[\mathbf{W}_{-a,k}, \mathbf{z}_k]$. If $M_{\text{UE}} < M_{\text{tx}}$, the adversary recovers only the projection of $\mathbf{R}_{\widetilde{\mathbf{x}}_k}$ onto the row space of $\widehat{\mathbf{H}}^{\text{adv}}_{a,k}$, and the reconstructed beampattern is distorted, which further favors the defender. The EM routine of [12] jointly refines the channel and signal estimates within this per-AP linear model. It does not resolve the inter-AP superposition, which is removed by the genie.

We have also rewritten the sentence explaining why the attack operates on raw samples, since in the original version it could be read as implying that raw samples alone suffice for per-AP recovery. In addition, Section IV now states that the simulated adversary is given accurate knowledge of $\mathbf{H}_{a,k}$, and that the reported success rates therefore correspond to a conservative attack model.

Location in the revised manuscript: Section II, threat-model paragraph; Section IV-A, simulation setup; Section IV-B, first paragraph. [Add page and line numbers.]

Revised text:

Section II: "One served user, indexed by a, is treated as an internal adversary. It knows its own pilots, data symbols, combiner $\mathbf{u}_a$, and the transmit-AP positions, and can form its own estimate $\widehat{\mathbf{H}}_{a,k}^{\mathrm{adv}}$ of its channel $\mathbf{H}_{a,k}$. Since all APs transmit common symbols, $\mathbf{y}_a[n]$ offers only M_{UE} dimensions for $N_{\mathrm{tx}}M_{\mathrm{tx}}$ per-AP transmit dimensions, so its per-AP decomposition is not identifiable from $\mathbf{y}_a[n]$ alone in general. Following [12], we therefore adopt a genie-aided strong-adversary model in which the per-AP components of the received samples are available to the adversary, as may occur when AP-specific transmissions are resolvable in time, frequency, or delay. Since the network cannot verify the receiver capability of a served user, this conservative model avoids relying on the adversary being unable to separate them. The attack operates on raw multi-antenna samples, since applying $\mathbf{u}_a$ would collapse each snapshot into a scalar. After decoding its own data symbol $s_a[n]$, the adversary subtracts its data component using its channel estimate $\widehat{\mathbf{H}}_{a,k}^{\mathrm{adv}}$, which leaves $\widetilde{\mathbf{y}}_{a,k}[n] = \widehat{\mathbf{H}}_{a,k}^{\mathrm{adv}}\widetilde{\mathbf{x}}_k[n] + \widetilde{\mathbf{n}}_{a,k}^{\mathrm{adv}}[n]$ with the non-self transmit component $\widetilde{\mathbf{x}}_k[n] = \sum_{j\neq a}\mathbf{w}_{j,k}s_j[n] + \mathbf{w}_{t,k}s_t[n]$, and $\widetilde{\mathbf{n}}_{a,k}^{\mathrm{adv}}[n]$ collecting the receiver noise and the residual caused by adversary-side channel-estimation errors. It then applies an expectation-maximization routine to obtain $\widehat{\mathbf{x}}_k[n] \approx \widetilde{\mathbf{x}}_k[n]$, which is identifiable when $\widehat{\mathbf{H}}_{a,k}^{\mathrm{adv}}$ has full column rank, i.e., with probability one for $M_{\mathrm{UE}} \geq M_{\mathrm{tx}}$, and forms the AP-dependent beampattern from the sample covariance over the available snapshots."

Comments by Reviewer 3

Comment 1:

Most of all, the legitimate sensing operation is not clearly described. For example, although multiple RX APs are considered, the operation at each RX AP (e.g., receive beamforming) and the combining between APs (coherent/non-coherent, central/distributed) are not described. It is difficult to grasp how the sensing SINR in Eq.2 is obtained. Also, it is not clear who experiences the sensing SINR. Is it an individual RX AP, or the system as a whole?

Response:

We thank the reviewer for this comment and agree that the legitimate sensing operation was not described in sufficient detail. The revised Section II now describes the processing chain as follows.

1. *Operation at each RX AP.* Each receive AP employs M_{rx} antennas and steers its receive beam toward the target. The resulting receive array gain is absorbed, together with the target RCS, in the scattering power $\xi_{r,k}^2$. Each receive AP then matched-filters its received samples with the known transmitted symbols. The data symbols are available at the central processor and therefore act as additional probing signals, while the echo of the known AN symbol is separated by matched filtering and discarded.

2. *Combining between RX APs.* The receive APs forward their matched-filter outputs to the central processor, which fuses them non-coherently. Non-coherent fusion avoids phase synchronization among the receive APs.

3. *How (2) is obtained.* At receive AP r, the echo is the superposition

$$\sum_{k=1}^{N_{\mathrm{tx}}} \alpha_{r,k}\sqrt{g_{r,k}}\,\mathbf{a}^H(\phi_k)\mathbf{W}_k\mathbf{s}[n],$$

   where $\mathbf{s}[n]$ collects the unit-power symbols and $\alpha_{r,k}$ is the bistatic echo coefficient with $\mathbb{E}\{|\alpha_{r,k}|^2\} = \xi_{r,k}^2$. Since the echo phases are not known to within the carrier wavelength, they are modeled as independent and uniformly distributed across transmit–receive AP pairs. Averaging over these phases and the symbols gives the per-snapshot SNR at receive AP r,

$$\gamma_r = \frac{1}{\sigma_{\mathrm{rx}}^2}\sum_{k=1}^{N_{\mathrm{tx}}} \xi_{r,k}^2 g_{r,k}\left\|\mathbf{a}^H(\phi_k)\mathbf{W}_k\right\|_F^2,$$

   and (2) is its average $\gamma_t = \frac{1}{N_{\mathrm{rx}}}\sum_r \gamma_r$.

4. *Who experiences the SNR.* γ_t is the average per-receive-AP SNR. The SNR of the fused statistic at the central processor is $N_{\mathrm{rx}}\gamma_t$. Since the two differ by a constant factor, they are maximized by the same beams, and the design is unaffected by this choice. All reported sensing SNRs are average per-receive-AP values, which is now stated in Section IV.

The detector applied to the fused statistic is described in our response to Comment 2.

Location in the revised manuscript: Section II, sensing-model paragraph; Section IV-A, simulation setup. [Add page and line numbers.]

Revised text:

> *Section II:* "Each receive AP employs M_{rx} antennas, steers its receive beam toward the target, and matched-filters its received samples with the known transmitted symbols, where the data symbols, being available at the central processor, act as additional probing signals. The bistatic echo phases are not known to the network to within the carrier wavelength and vary with the aspect angle, and are therefore modeled as independent and uniformly distributed across transmit–receive AP pairs. Averaging over these phases removes the cross terms between the echoes of different transmit APs. With $\mathbf{W}_k \triangleq [\mathbf{w}_{1,k}, \ldots, \mathbf{w}_{N_{\mathrm{UE}},k}, \mathbf{w}_{t,k}]$, the per-snapshot sensing SNR, averaged over the echo phases and the N_{rx} receive APs, is [(2), unchanged], where ..., $\xi_{r,k}^2$ is the scattering power absorbing the target radar cross section (RCS) and the receive array gain, and σ_{rx}^2 is the receive noise power. The receive APs forward their matched-filter outputs to the central processor, which fuses them non-coherently, so the fused SNR $N_{\mathrm{rx}}\gamma_t$ is maximized by the same beams as γ_t."

Comment 2:

Furthermore, the authors have not described any detector or detection threshold. Therefore, "x dB sensing SINR loss" doesn't represent anything. How is it translated into the detection and false-alarm probabilities, which eventually matter? Without this, it is very difficult to judge whether the proposed scheme is practically useful.

Response:

We thank the reviewer for this important comment and agree that the SNR loss should be related to detection performance. The revised manuscript now specifies the detector and reports the corresponding detection probabilities.

Detector. Each receive AP matched-filters its samples with the $N_{\mathrm{s}} = N_{\mathrm{UE}} + 1$ known data and sensing streams. With unknown echo amplitudes and phases, the generalized likelihood ratio test sums the matched-filter output energies over the N_{s} streams and the N_{rx} receive APs, and compares the sum with a threshold τ. After normalization, the statistic follows a central χ^2 distribution with $2N_{\mathrm{rx}}N_{\mathrm{s}}$ degrees of freedom under the null hypothesis, so τ is set by $\Pr\{\chi^2_{2N_{\mathrm{rx}}N_{\mathrm{s}}} > \tau\} = P_{\mathrm{fa}}$. Under the target hypothesis, it follows a noncentral χ^2 distribution with noncentrality $2N\sum_r \gamma_r$, which gives

$$P_{\mathrm{d}} = \mathbb{E}\Big\{Q_{N_{\mathrm{rx}}N_{\mathrm{s}}}\Big(\sqrt{2N\textstyle\sum_r \gamma_r}, \sqrt{\tau}\Big)\Big\},$$

where $Q_M(\cdot,\cdot)$ is the generalized Marcum Q-function of order M. Consistent with the random echo phases of the sensing model, the per-receive-AP SNRs γ_r are taken as independent and exponentially distributed with mean γ_t, and the expectation is over this fluctuation. Among fluctuation models with the same mean, this is a conservative choice in the high-P_{d} regime.

Detection performance. For $P_{\mathrm{fa}} = 10^{-6}$, $N_{\mathrm{rx}} = 4$, and $N_{\mathrm{s}} = 5$, $P_{\mathrm{d}} = 0.9$ requires $\gamma_t \geq -8.5$ dB with integration over the $N = 128$ snapshots. The proposed design at $\eta \approx 0.50$ ($\gamma_t = 11.1$ dB) therefore retains a margin of 19.6 dB, compared with 23.2 dB for the no-AN reference ($\gamma_t = 14.7$ dB), and both designs achieve $P_{\mathrm{d}} > 0.999$. Even with the shortest

integration that resolves all streams, $N = N_{\mathrm{s}} = 5$, the proposed design achieves $P_{\mathrm{d}} = 0.998$, compared with 0.9999 for the no-AN reference. The 3.6 dB loss therefore reduces the detection margin rather than the detection probability in the default setting.

Practical interpretation. Since the two-way path gain $g_{r,k}$ scales with the fourth power of distance, a 3.6 dB loss corresponds to an approximately 19% reduction of the bistatic distances that can be supported at fixed $(P_{\mathrm{d}}, P_{\mathrm{fa}})$. Moreover, as stated in our response to Reviewer 2, Comment 1, the reported loss is an upper bound, since the known AN echo is discarded. Given the length limit of a correspondence, we report these values in the text of Section IV-B rather than in an additional figure.

Location in the revised manuscript: Section IV-B, discussion of Fig. 3. [Add page and line numbers.]

Revised text:

> *Section IV-B:* "To relate these values to detection performance, we consider the generalized likelihood ratio test at the central processor, which sums the matched-filter output energies over the $N_{\mathrm{UE}} + 1$ known streams and the N_{rx} receive APs, with the threshold set for $P_{\mathrm{fa}} = 10^{-6}$. With per-receive-AP SNRs exponentially distributed around γ_t due to the random echo phases, $P_{\mathrm{d}} = 0.9$ requires $\gamma_t \geq -8.5$ dB over $N = 128$ snapshots, so the proposed design at $\eta \approx 0.50$ retains a 19.6 dB margin, compared with 23.2 dB without AN, and both achieve $P_{\mathrm{d}} > 0.999$. The 3.6 dB loss thus reduces the detection margin rather than the detection probability, and corresponds to an approximately 19% reduction of the supported bistatic distances at fixed $(P_{\mathrm{d}}, P_{\mathrm{fa}})$."

Comment 3:

[Insert the reviewer's third comment verbatim.]

Response:

[Provide the response and describe any resulting changes.]

Location in the revised manuscript: [Section, page, and line numbers, or not applicable.]

Revised text:

> [Insert the relevant revised passage, if applicable.]

Comment 4:

[Insert the reviewer's fourth comment verbatim.]

Response:

[Provide the response and describe any resulting changes.]

Location in the revised manuscript: [Section, page, and line numbers, or not applicable.]

Revised text:

[Insert the relevant revised passage, if applicable.]

Comment 5:

Minor comments.

- RCS is not specified in the experiments.
- Eq. 2 is not really an SINR. No interference there.
- TheRX-AP antenna number is missing.

Response:

We thank the reviewer for these careful observations, which we have addressed as follows.

- *RCS.* The target RCS used in the simulations, $\sigma_{\mathrm{RCS}} = $ [value] m^2, is now listed in Table I, and Section II states that $\xi_{r,k}^2$ absorbs the RCS and the receive array gain.
- *SNR versus SINR.* We agree. Since the data symbols are known probing signals and the AN echo is separated by matched filtering, (2) contains no interference term and is an SNR. The revised manuscript uses the term "sensing SNR" consistently and states that the AN echo is separated and discarded.
- *RX-AP antenna number.* The number of receive-AP antennas, $M_{\mathrm{rx}} = $ [value], is now defined in Section II and listed in Table I.

Location in the revised manuscript: Section II, sensing-model paragraph; Table I. [Add page and line numbers.]

Revised text:

Table I (added rows): "Target RCS σ_{RCS}: [value] m^2; RX-AP antennas M_{rx}: [value]."